

OKAMOTO IDENTIFICATION PROTOCOL:-

Okamoto Identification protocol is a cryptographic interactive protocol designed for proving knowledge of two secret exponents related to a public key constructed in a group with two generators.

It is a generalization of Schnorr's identification scheme, allowing one to prove possession of a pair of secrets without revealing them.

Let G be a cyclic group of prime order q with two generators g and h publicly known.

Let $(\alpha, \beta) \in \mathbb{Z}_q^2$ be a secret key (witness) pair

and $X = g^\alpha h^\beta \in G$ be the public key

This public key binds the secret exponents to the group elements.

The interactive process between a prover P who knows (α, β) and a verifier V who knows X proceeds as follows:

1) COMMITMENT :

- Prover selects random values $k_\alpha, k_\beta \in \mathbb{Z}_q$
- Computes the commitment $R = g^{k_\alpha} h^{k_\beta}$
- Sends R to the verifier.

2) CHALLENGE :

- Verifier selects a random challenge $e \in \mathbb{Z}_q$
- Sends e to the prover.

3) RESPONSE :

- Prover computes

$$s_\alpha = k_\alpha + e_\alpha \pmod{q_1}$$

$$s_\beta = k_\beta + e_\beta \pmod{q_2}$$

- Sends the response pair (s_α, s_β) to the verifier.

4. VERIFICATION:

- Verifier checks whether $g^{s_\alpha} h^{s_\beta} \stackrel{?}{=} R \cdot X^e$
- Accepts if equality holds, rejects otherwise.